

E-commerce Customer Purchase & Retention Analysis

Objective

The objective of this project is to analyze customer purchasing behavior and retention patterns in an e-commerce platform.

This analysis focuses on:

- Understanding customer purchase frequency
- Identifying repeat vs one-time customers
- Analyzing revenue trends
- Evaluating category-wise sales performance
- Identifying high-value customers

Dataset Description

The dataset consists of 3 tables:

Customers Table

- customer_id
- signup_date
- city

Orders Table

- order_id
- customer_id
- order_date
- total_amount

Order Items Table

- item_id
- order_id
- product_name
- category
- price



Total Data

- 500 customers
- 1000 orders
- 3000+ order items

SQL Analysis

The analysis was performed using SQL queries involving:

- JOIN operations → combining multiple tables
- Aggregation functions → SUM(), AVG(), COUNT()
- GROUP BY & ORDER BY → segmentation
- CASE statements → customer classification

Queries

- Repeat vs New Customers

```
SELECT  
CASE  
WHEN COUNT(order_id) = 1 THEN 'One-time'  
ELSE 'Repeat'  
END AS customer_type,  
COUNT(*) AS total_customers  
FROM orders  
GROUP BY customer_id;
```

- Top Customers (Revenue)

```
SELECT customer_id, SUM(total_amount) AS total_spent  
FROM orders  
GROUP BY customer_id  
ORDER BY total_spent DESC  
LIMIT 10;
```



- Monthly Revenue

```
SELECT
  DATE_FORMAT(order_date, '%Y-%m') AS month,
  SUM(total_amount) AS revenue
FROM orders
GROUP BY month
ORDER BY month
```

- Average Order Value

```
SELECT ROUND(AVG(total_amount),2) AS avg_order_value
FROM orders;
```

- Category-wise Sales

```
SELECT category, SUM(price) AS total_sales
FROM order_items
GROUP BY category
ORDER BY total_sales DESC;
```

Key Insights

Add this section (your previous file was missing strong insights)

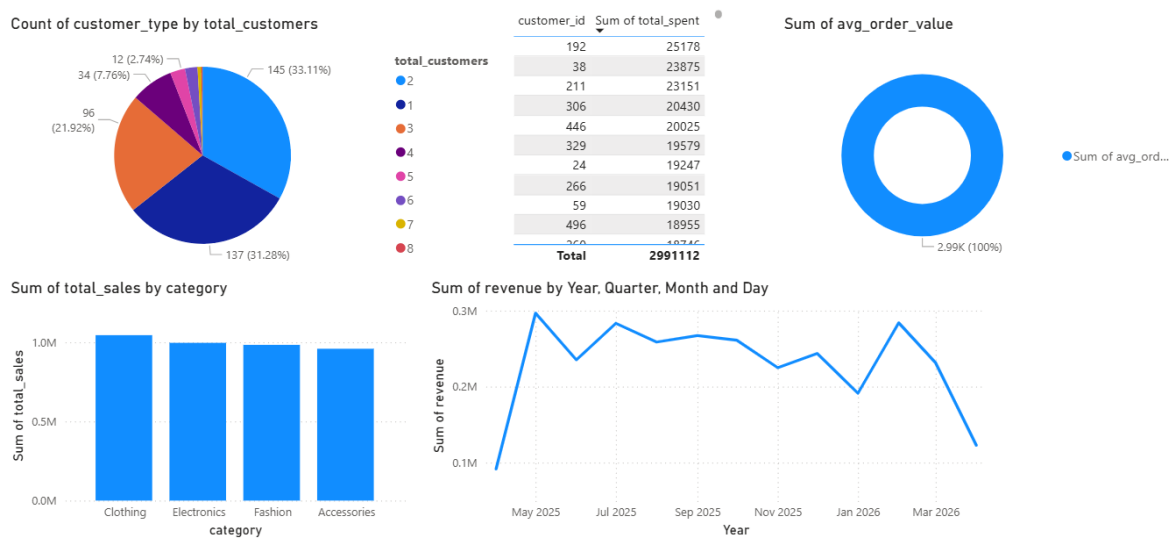
- A small number of customers contribute the majority of revenue
- Repeat customers generate higher total spending compared to one-time buyers
- Electronics and Fashion categories contribute the highest sales
- Monthly revenue shows consistent growth with minor fluctuations
- Average order value indicates mid-range purchasing behavior



Dashboard

An interactive dashboard was created using Microsoft Power BI to visualize insights:

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Conclusion

This project demonstrates how customer behavior analysis can help improve business decisions.

The analysis shows that:

- Retaining existing customers is more profitable than acquiring new ones
- High-value customers play a critical role in revenue generation
- Category performance helps in optimizing inventory and marketing strategies

